The Journal of Open Source Software

# LinguiPhyR: A Package for Linguistic Phylogenetic Analysis in R

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**DOI:** 10.xxxxxx/draft

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**Editor:** Andrew Stewart

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**Submitted:** 29 November 2023
**Published:** unpublished

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## Introduction

Phylogenetic methods have become commonplace in historical linguistics research. However, much of the work is highly technical and not easily accessible to the typical classically-trained historical linguist. This paper aims to bridge the gap between linguistic and statistical research by introducing LinguiPhyR, an R package that provides a graphical user interface (GUI) to aid in the phylogenetic analysis of linguistic data. As such, very little computational background is required by the user. A linguist may simply upload a dataset, select optimization criteria, and visualize the phylogenies found by the search algorithm. Alternatively, one may upload trees of interest to be analyzed based on the dataset. Several tools for tree analysis are provided: users may examine what characters are responsible for particular splits in the tree, see the characters that are incompatible on the tree, annotate internal nodes of the tree with reconstructed states, and even see a relative chronology of state changes.

We note that, at present, our software focuses on parsimony-based tree estimation and analyses. We make this choice because such an approach is easily interpretable: the best tree is simply the tree that minimizes the number of state changes. This makes it easy for linguists to see the effect of each character in the dataset on tree search. However, one limitation of parsimony-based methods is that they are limited to searching for and analyzing tree topology: studies seeking to explore ancestral node dating (glottochronology) or branch lengths are better suited to using likelihood or Bayesian approaches. Future work will include the incorporation of other search algorithms and analytical methods into LinguiPhyR.

## Statement of need

Given the recent explosion of new linguistic phylogenetic datasets (Heggarty et al., 2023; Herce & Cathcart, 2024; Jäger, 2018; Tresoldi, 2023), new tools for their analyses are called for. Many linguists want to perform parsimony analyses of their dataset, and our software makes it easy to do so with little effort. In this work, we provide an easy-to-use parsimony-based tool for phylogenetic analysis that emphasizes *interpretability*, allowing linguists to understand why trees are returned for a particular dataset *or* what evidence a new dataset has for existing trees suggested by the community. Currently, the go-to method for phylogenetic analysis is Bayesian inference, which, despite efforts to reduce barrier to entry, requires reasonable mathematical maturity to understand and operates largely as a black-box.

The primary goals of LinguiPhyR are to

1. Make phylogenetics accessible to linguists by requiring *no* coding or writing of configuration files. While these are useful skills, giving linguists the option to spend their time analyzing trees in a GUI rather than writing code will facilitate analyses of phylogenetic inferences.

2. Make it easy to find and visualize trees for a new linguistic dataset. One simply has to

upload the dataset and select optimization criteria (or use the default settings). Trees are then displayed in the app and can be downloaded (either as images or as Nexus files) for inclusion in other work.

3. Provide a comprehensive set of (parsimony-based) analysis tools. These focus on the following questions: why are particular trees being suggested for the dataset? What evidence does a dataset contain for other trees of interest? What is the effect of particular coding decisions in the dataset on the understanding of a tree?

Our work is not the only attempt to make phylogenetic methods accessible and interpretable to linguists, nor is it the only GUI for this purpose. For example, PAUP* (Swofford, 2002) provides a GUI containing a comprehensive set of parsimony-based tools for phylogenetics, although it does require writing Nexus configuration files and is not specifically aimed at linguists. Tools specific to Bayesian linguistic phylogenetics include BEASTling (Maurits et al., 2017), which is a wrapper for BEAST (Bouckaert et al., 2014), and Traitlab (Kelly et al., 2023). A useful tutorial in R for linguistic phylogenetics is Goldstein (2020).

# LinguiPhyR: Linguistic Phylogenetic Analysis in R

The following sections describe each page of the app: Data Upload, Tree Search, and Analysis. Throughout the subsequent discussion, many terms familiar to historical linguists are used (e.g. *clade*, *cognate*, and *regular sound change*); we suggest Ringe & Eska (2013) for further reading. Similarly, we recommend Warnow (2017) for terms common in the phylogenetics literature, such as *character*, *polymorphism*, and *parsimony*.

## Data Upload

The user first uploads a dataset of linguistic characters, which encode certain properties about languages that are likely to be relevant to the branching structure of the underlying tree. The characters should be specified in a spreadsheet and uploaded as a CSV file. An example of the data format is shown below[1]:

**Table 1:** Example dataset specification, excerpted from the screened Indo-European dataset of Ringe et al. (2002).

| id | feature | weight | char-type | HI | AR | GK | AL | TB | VE | AV | OC | LI | ⋯ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| c1 | P1 | 50 | standard | 4 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | ⋯ |
| c26 | M3 | 50 | standard | 1 | 2 | 2 | 3 | 2 | 2 | 2 | 2 | 4 | ⋯ |
| c50 | bird | 1 | standard | 1 | 2 | 3 | 4 | 5 | 6 | 6 | 7 | 8 | ⋯ |

Each row represents a character. The first four columns specify special character information: a unique character ID, the character name ("feature"), the weight of the character (optional, to be used in parsimony analyses), and the character type (which can be *standard*, *irreversible*, or *custom*, explained below). The remaining columns contain the character states for each attested language (i.e. the leaves of the tree).

Two languages should be given the same state for a character *if and only if* the languages' realization of that character could be from a common genetic source (and not, for example,

[1] We provide the screened version of the Indo-European dataset of Ringe et al. (2002) in the correct format at the path `data/ringe_screened_dataset.csv` in the LinguiPhyR Github repository.

from borrowing). For lexical data, characters typically represent particular semantic slots (such as "bird" in the table above), and languages should share a state if their words for that meaning are cognate — that is, the words are derived from a common ancestor via regular sound change. However, if a linguist can demonstrate that two languages share the same cognate due to borrowing or some other non-genetic source, then the languages should be given different states for that character.

Such cognate judgements are critically important to the results of phylogenetic estimation. A haphazard or automated data representation will not yield meaningful trees; hence, it is important to have well-trained linguists judge relevant material and select characters that actually represent potentially shared innovations. An abundance of phylogenetics literature discusses good methodology for doing this Heggarty (2021); classical historical linguistics references are also helpful Campbell (2013). Further, our coding scheme is applicable to phonological, morphological, and structural/typological characters, which are abundant in phylogenetic datasets.

Each character may be declared "standard", "irreversible", or "custom". Standard characters permit any change of state (e.g. from 0 to 1 or from 1 to 2) with uniform cost. This is generally appropriate for lexical characters where the states represent cognate classes. Irreversible characters are binary characters that may transition from 0 to 1 but not from 1 to 0. This is appropriate in the case of phonological mergers, which are generally considered irreverisble. Finally, custom characters allow the user to declare which state transitions are allowed, and what the cost should be for each permitted transition. The exact way to specify this is described in the "Data Upload" page of the app.

Our data format also supports *polymorphic* character states: these are instances where a language exhibits more than one state for a character. In the context of lexical data, this would mean that a language manifests two cognate classes for the same semantic slot. Such examples are denoted by separating the states with a / (e.g. *1/2*) in the dataset.

Finally, we note that our software permits *multi-state* characters, not just binary traits. Binary traits are particularly common in likelihood-based phylogenetic estimation, because most likelihood models require a pre-specified state space (e.g. $0$ and $1$). Non-parametric methods like parsimony do not have this assumption, and, in fact, it is not advisable to treat multi-state characters as a set of binary traits because the estimation algorithms consider the traits independent (when they are not) (Nichols & Warnow, 2008; Rexová et al., 2003; Warnow, 2017). For example, a lexical character denoting "bird" may have states $1$, $2$, and $3$, each representing a different cognate class observed in attested languages. Treating this as binary would create three traits, referring to whether or not the languages exhibit each of these cognate classes in the "bird" meaning. Unless there is a reason to make this binary conversion (e.g. because it is necessary to run likelihood algorithms), we suggest to leave the data in the underlying multi-state form.

The app then presents some statistics about the dataset, and one can perform some simple analyses:

- **Parsimony Uninformative Characters:** The characters that are not *parsimony informative* are displayed. These characters will have no effect on parsimony-based tree estimation because they can be fit equally well to any tree (see Warnow (2017) for a discussion). This is especially helpful to a linguist, who may not be thinking about the consequences of character codings to the parsimony algorithm when coding individual characters. This thus allows a linguist to carefully consider coding choices.

- **Character-level Statistics:** Various information about each character is displayed, such as the number of languages having polymorphic states for that character and whether or not the character is parsimony-informative (among others). The dataset may be sorted by these metrics.

- **Clade Analysis:** The user may select a subset of languages and analyze what characters

provide support for such a clade (a clade is a subset of languages separated from all other languages by an edge in the tree). This is computed in the strictest sense: a character only supports a hypothetical clade if the languages in the clade all share the same state, and all other languages share a different state.[2]

An example usage of the "Data Upload" page is shown in Figure 1; here one can see the screened dataset of Ringe et al. (2002) uploaded.

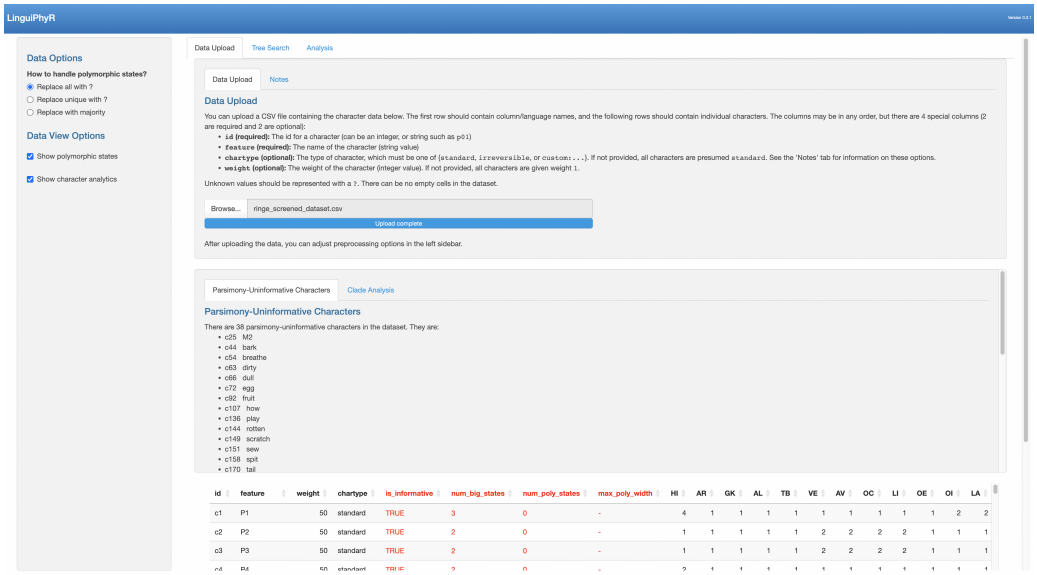


**Figure 1:** "Data Upload" page of LinguiPhyR.

## Tree Search

Then, the user may proceed to the second page of the app, which conducts a search for the optimal tree(s) given the dataset. We use PAUP* (Swofford, 2002) to perform tree search, a well-established package in the biological community for running parsimony and other phylogenetic analyses. The user may specify various optimization criteria in the app without having to write configuration files by hand, which is a big barrier to entry for many linguists. Nonetheless, users may download these configuration files from the app and modify them as needed.

Figure 2 demonstrates tree search using PAUP*.

[2]It is important to note that a clade *on a particular tree* may be supported by more than just the characters that meet this condition. For example, if the dominant cognate class in a clade is lost by just one language in the clade, the character will still support the grouping if the removal of the edge separating the clade from all other languages would produce a less parsimonious tree. This can be examined in the "Analysis" page of the application.

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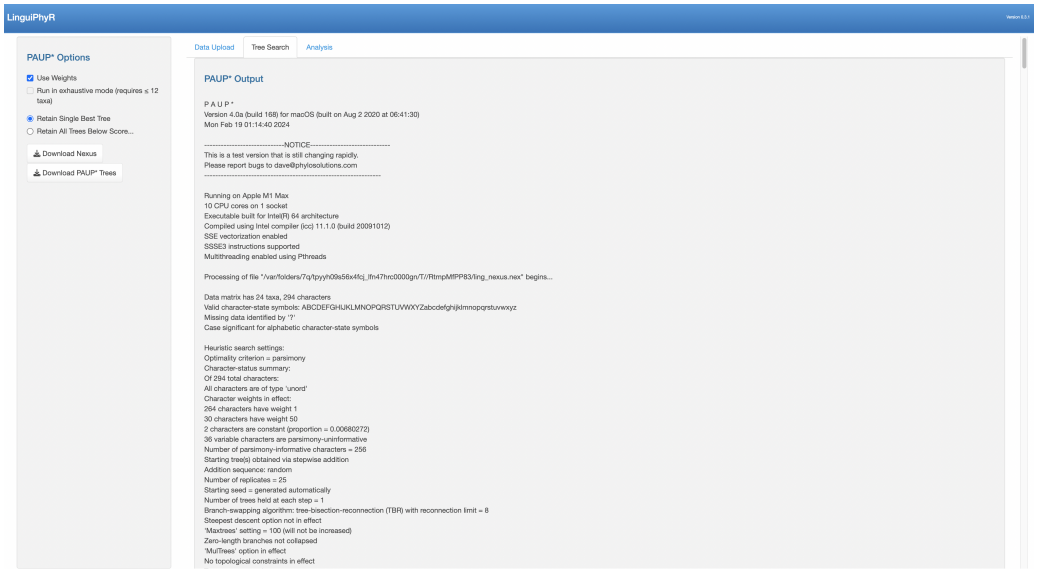


**Figure 2:** "Tree Search" page of LinguiPhyR.

## Analysis

Finally, one may use the dataset to analyze trees. These trees can be either the result of a PAUP* tree search, or specific trees of interest uploaded by the user. This latter option is especially helpful for determining the support that a dataset exhibits for various trees accepted by the community. Strict and majority consensus trees for the trees returned by PAUP* are displayed as well. The primary analyses that can be performed on a tree are the following:

1. **Tree Score:** Each tree is scored using various metrics, including *parsimony*, *compatibility*, *total edge support*, and *minimum edge support*[3]. Hence, the trees can be ranked according to these options.

3. **Character annotations:** The user may select any character and see the most parsimonious annotation(s) of that character's states across the tree (including reconstructed states at internal nodes). This is convenient for studying a character's behavior, and can help a linguist interpret the consequences of particular character codings on phylogeny estimation. For example, when coding the absence of a feature, a linguist has two choices: either code all languages without the feature with the same state (e.g. $0$), or code them all with different states. The former choice would suggest the absence of feature as evidence of a clade among those languages without the feature, while the latter would suggest that absence of the feature is not evidence that the languages are related. By annotating the states of each choice on proposed trees, the linguist can see the most parsimonious evolution patterns for both codings.

4. **Incompatible characters:** This reports the characters that are not compatible on a tree. This is useful for considering how plausible various trees are: if the set of characters that a tree is not compatible on seems unrealistic, a linguist may wish to discard the tree in favor of other options.

5. **Enforcing characters:** This reports the characters that enforce, or support, each edge in the tree. A character is deemed to support an edge if and only if the edge's collapse

[3]The compatibility score is the total number of characters that evolve on the tree without homoplasy (see Warnow (2017) for further detail). To calculate total edge support and minimum edge support, we first calculate the number of characters that enforce, or support, each edge, based on whether or not the collapse of that edge would increase the parsimony score. Total edge support is the sum of these support values across all edges, and minimum edge support is the minimum of these values.

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increases the parsimony score for that character. This feautre allows one to analyze evidence for and against various clades.

6. **Relative chronology:** This reports a relative chronology of state changes *across* characters. This is calculated by first determining the most parsimonious state transitions for each character, and then ordering these transitions based on the edges they occur on from the root of the tree to a specified clade. This type of relative chronology may seem unusual to the typical historical linguist, but its results can be illuminating.

Figure 3 depicts an example tree analysis in LinguiPhyR, based on the screened dataset of Ringe et al. (2002).

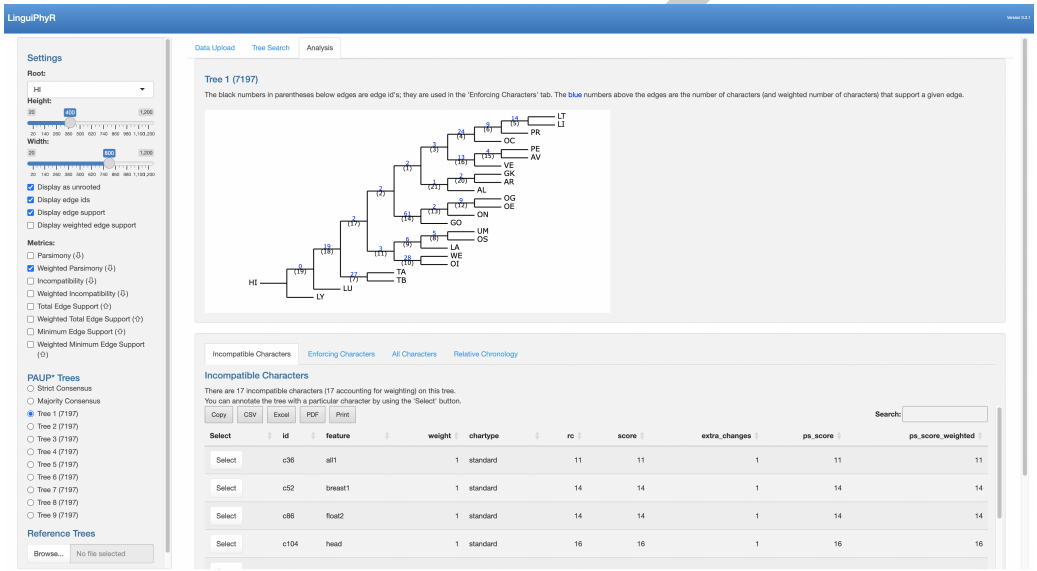


**Figure 3:** Analysis page of LinguiPhyR.

# Conclusions

We present LinguiPhyR, a useful tool for analyzing phylogenetic datasets and trees without the need to code. Even for experienced programmers, LinguiPhyR can quickly enable analysis on a new linguistic dataset or provide a starting place for finding new trees. In our app, we especially emphasize parsimony-based interpretability by providing useful visualizations and tools to see the impact of certain coding decisions on tree estimation. Future work will include the incorporation of other inference methods (such as distance-based and quartet approaches), as well as more advanced analytical tools, such as bootstrap analysis.

# Acknowledgements

The author would like to acknowledge Thomas Olander, Matthew Scarborough, Simon Poulsen, Anders Jørgensen, Stefanos Baziotis, and Tandy Warnow, who have all provided invaluable feedback throughout the project.

This research was supported by the research project Connecting the Dots: Reconfiguring the Indo-European Family Tree (2019–2024), financed by the Independent Research Fund Denmark (project number 9037-00086B).

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